



ACTL3142 Tutorial

Week 1: Introduction to Statistical Learning

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With references to Tadhg Xu-Glassop's slides
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Introduction

Hi! My name is **Nick** and I'm excited to be tutoring you this term.

A little about me:

- Fourth-year Actuarial Studies and Computer Science student
- Interested in poker and basketball
- I took this course in T2 2024

Advice

This course is far too rich to learn exclusively within tutorials. Please keep up with lectures and ideally read the website and textbook. The exam is typed (Inspira), so focus on **intuition and application** rather than memorising formulas.

Icebreakers

Please go around and introduce yourself:

1. What's your name?
2. Year and degree
3. Favourite course you've taken
4. An emoji to describe your day so far

What is Statistical Learning?

A vast set of tools for **understanding data** and **predicting outcomes**.

We believe that data follows a model of the form:

$$Y = f(\mathbf{X}) + \varepsilon$$

- f is some unknown function that captures the **systematic** (deterministic) relationship between inputs and output
- $\mathbf{X} = (X_1, \dots, X_p)$: the predictors (features, independent variables)
- Y : the response (output, dependent variable)
- ε : the irreducible error, i.e. everything $f(\mathbf{X})$ cannot explain (random chance, measurement error, missing variables, etc.)

Our goal: estimate f . The resulting $\hat{f}$ is our **model**.

Prediction vs. Inference

Prediction

- Goal: accurately estimate Y given X
- The form of f is not the focus; we just want accurate outputs
- Models can be very complex (black box is fine)

Inference

- Goal: understand *how* X affects Y
- Which predictors matter? What is their individual effect?
- Models tend to be simpler and more interpretable

Key questions to always ask

What predictors should we include? Do we want to predict or explain? What is signal versus noise?

Key Distinctions

- **Supervised** (Y observed) vs. **Unsupervised** (no Y ; find structure)
- **Regression** (Y quantitative) vs. **Classification** (Y categorical)
- **Interpretability** vs. **Flexibility**: easy to understand f or able to capture complex relationships (typically a tradeoff)

Two approaches to estimating f :

Parametric

- Assume a form for f (e.g. linear)
- Estimate a few parameters
- Works with limited data
- Risk: misses signal (**high bias**)

Non-parametric

- No assumption about f 's shape
- Many effective “parameters”
- Needs lots of data
- Risk: fits noise (**high variance**)

CQ2: Classification or Regression? (ISLR2 Q2.2)

2. (ISLR2, Q2.2) Explain whether each scenario is a classification or regression problem, and indicate whether we are most interested in inference or prediction. Finally, provide n and p .
- a. We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.
 - b. We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a **success** or **failure**, price charged for the product, marketing budget, competition price, and ten other variables.
 - c. We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.

Loss Functions and the Bias-Variance Tradeoff

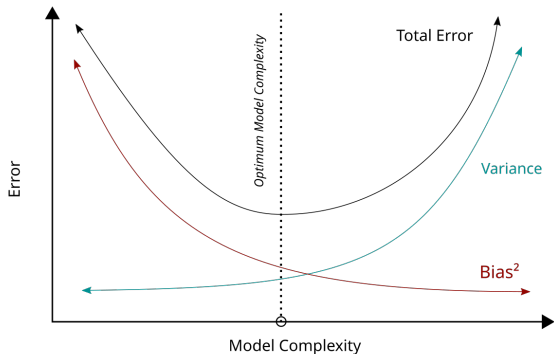
Suppose our model predicts $\hat{y}$. We compare this to the true y using a **loss function** (e.g. squared error).

The expected test MSE decomposes as:

$$\underbrace{E[(y_0 - \hat{f}(x_0))^2]}_{\text{Expected Test MSE}} = \underbrace{[\text{Bias}(\hat{f}(x_0))]^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{Variance}} + \underbrace{\text{Var}(\varepsilon)}_{\text{Irreducible}}$$

- **Bias**: systematic error from simplifying assumptions
- **Variance**: how much $\hat{f}$ changes across different training sets
- $\text{Var}(\varepsilon)$: irreducible error, a floor we cannot beat

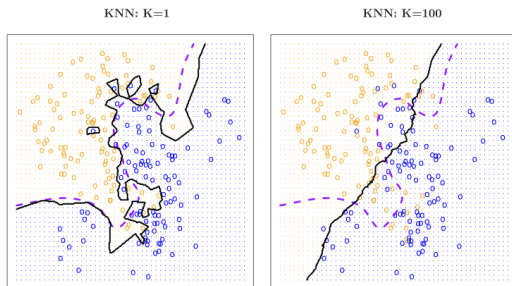
Visualising the Tradeoff



- More flexible $\Rightarrow$ less bias, more variance
- Less flexible $\Rightarrow$ more bias, less variance
- The best model achieves an **optimal tradeoff** at the minimum of total error

Bias-Variance in Action: KNN

KNN classifier: to classify a new point, find its K nearest neighbours and let them vote. The majority class wins.



$K = 1$ (very flexible)

Low bias, high variance $\rightarrow$ **overfitting**

$K = 100$ (inflexible)

High bias, low variance $\rightarrow$ **underfitting**

Some K between 1 and 100 achieves a better bias-variance tradeoff.

Simple Linear Regression: Let's Work Through an Example

A **parametric** approach. We assume a structure for f :

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Interpreting the coefficients:

- β_0 (intercept): the predicted value of y when $x = 0$
- β_1 (slope): the average change in y for a one-unit increase in x

Properties:

- Not very flexible, since we impose a linearity assumption
- But easy to interpret: a clear example of the flexibility-interpretability tradeoff

Estimating the coefficients:

- Given training data, we need assumptions about ε to estimate β_0 and β_1
- The idea: pick the β 's that **minimise the loss function**
- Here we use **MSE** (Mean Squared Error) as our loss

CQ1: Flexible vs. Inflexible Methods (ISLR2 Q2.1)

1. ★ (ISLR2, Q2.1) For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer.
 - a. The sample size n is extremely large, and the number of predictors p is small.
 - b. The number of predictors p is extremely large, and the number of observations n is small.
 - c. The relationship between the predictors and response is highly non-linear.
 - d. The variance of the error terms, i.e., $\sigma^2 = \mathbb{V}(\epsilon)$, is extremely high.

CQ3: Bias-Variance Decomposition (ISLR2 Q2.3)

3. ★ (ISLR2, Q2.3) We now revisit the bias-variance decomposition.
- Provide a sketch of typical (squared) bias, variance, training error, test error, and Bayes (or irreducible) error curves, on a single plot, as we go from less flexible statistical learning methods towards more flexible approaches. The x -axis should represent the amount of flexibility in the method, and the y -axis should represent the values for each curve. There should be five curves. Make sure to label each one.
 - Explain why each of the five curves has the shape displayed in part (a).

CQ4: Advantages and Disadvantages of Flexibility (ISLR2 Q2.5)

4. ★ (ISLR2, Q2.5) What are the advantages and disadvantages of a very flexible (versus a less flexible) approach for regression or classification? Under what circumstances might a more flexible approach be preferred to a less flexible approach? When might a less flexible approach be preferred?

Applied Questions

Time to switch to R and work through the applied exercises.

What we'll cover:

- Exploratory data analysis (EDA) with real datasets
- Using `summary()`, `pairs()`, `plot()`, and `hist()`
- Understanding your data *before* fitting any models

Key Takeaway

Always explore your data first. Summary statistics and visualisations reveal patterns, outliers, and potential issues that inform every modelling decision you make.

Thanks for coming! See you next week.

